

6.1.6 Lab: Connect Fiber Optic Cables

null

Candidate: omar santiago (osantiago162@gmail.com)**Time Spent:** 05:07**Score:** 100%

Task Summary

Required Actions

- ✓ Connect the router to the power strip
- ✓ Connect the router to the computer
- ✓ Connect the router to the ISP's fiber line

Explanation

Complete the following steps:

1. Add the router to the workspace and connect it to the data line.
 - a. In the Inventory, expand the **Networking Devices** category.
 - b. Drag the fiber router from the Inventory to the Workspace.
 - c. To connect the router to the Internet connection, switch to the back view of the fiber router.
 - d. In the Inventory, expand the **Cables** category.
 - e. Select the **SC to SC** cable.
 - f. In the Selected Component window, drag a connector to the port on the fiber router.
 - g. In the Selected Component window, drag the other connector to the port on the wall plate.
2. Connect the router to the computer.
 - a. Switch to the back view of the computer.
 - b. In the Inventory, select the Cat6a Ethernet cable.
 - c. In the Selected Component window, drag a connector to an Ethernet port on the fiber router.
 - d. In the Selected Component window, drag the other connector to the Ethernet port on the computer.
3. Connect the router to the power strip.
 - a. Select the power adapter in the Inventory.
 - b. In the Selected Component window, drag the DC power connector to the port on the fiber router.
 - c. In the Selected Component window, drag the AC adapter end to an outlet on the power strip.

Accessible Steps to Complete the Lab:

1. Add the Fiber Router to the workspace:

- a. Press **ALT-I** to access the **Inventory** menu.
 - b. Tab to **Networking Devices** and press **Enter**.
 - c. Tab to the **Fiber Modem/Router** and press **Enter**.
 - d. Press **Enter** again to **Add** the **printer** to the **workbench**.
 - e. On the **Select Location to Add Item** window:
 - i. In the drop-down menu under **Connector End to Connect**, press the **Down arrow** key until **Fiber Modem/Router, USB or 100/1000BaseTX Ethernet** is shown.
 - ii. Press **Tab** to access the **Destination** drop-down menu.
 - iii. Use the **Down arrow** key until **Bench** is shown and press **Enter**.
 - iv. Press **Tab** twice and then **Enter** to **Confirm**.
 - f. The **modem** should now be on the **workbench**.
2. **Connect the modem to the ISP's fiber line:**
- a. Press **ALT-I** to access the **Inventory** menu.
 - b. Tab to **Cables** and press **Enter**.
 - c. Tab to **Cable, Fiber, SC to SC** and press **Enter**. Press **Enter** again to **add** the **cable**.
 - d. On the **Select Location to Add Item** window:
 - i. In the drop-down menu under **Connector End to Connect**, press the **Down arrow** key until **Fiber Optic SC Connector (A)** is shown.
 - ii. Press **Tab** to access the **Destination** drop-down menu.
 - iii. Use the **Down arrow** key until **Fiber Port Fiber Modem** is shown and press **Enter**.
 - iv. Press **Tab** twice and then **Enter** to **Confirm**.
 - e. Press **Alt-W** to access the **Workspace**.
 - f. Tab until the arrow next to **Fiber Modem/Router, USB or 100/1000BaseTX Ethernet** is highlighted and press **Enter**.
 - g. Tab two times so the arrow next to **Fiber Optic SC Connector (A)** is highlighted and press **Enter**.
 - h. Tab one time so **Fiber Optic SC Connector (A)** is highlighted and press **Enter**. Press **Enter** again to **add** the **connector**.
 - i. In the **Select Location to Add Item Window**:
 - i. Tab to the **Destination** field and press **Enter**. Use the **down arrow** key until **Fiber Outlet, AC 115 volt plus SC** is selected and press **Enter**.
 - ii. Press **Tab** twice and then **Enter** to **confirm**.
3. **Connect the modem to the computer:**
- a. Press **Alt-I** and Tab to **Cables** and press **Enter**.
 - b. Tab to **Cat6a Cable, RJ45** and press **Enter**. Press **Enter** again to **add** the **cable**.
 - c. On the **Select Location to Add Item** window:
 - i. In the drop-down menu under **Connector End to Connect**, press the **Down arrow** key until **RJ45 Shielded Connector (568B)** is shown.
 - ii. Press **Tab** to access the **Destination** drop-down menu.
 - iii. Use the **Down arrow** key until **Ethernet 1 Fiber Modem** is shown and press **Enter**.
 - iv. Press **Tab** twice and then **Enter** to **Confirm**.
 - d. Press **Alt-W** to access the **Workspace**.
 - e. Tab until the arrow next to **Fiber Modem/Router** is highlighted and press **Enter**.
 - f. Tab two times so the arrow next to **RJ45 Shielded Connector (568B)** is highlighted and press **Enter**.
 - g. Tab one time so **Connector, Shielded, RJ45, 568B, Male** is highlighted and press **Enter**. Press **Enter** again to **add** the **connector**.
 - h. In the **Select Location to Add Item Window**:
 - i. Tab to the **Destination** field and press **Enter**. Use the **down arrow** key until **Network Motherboard, Socket AM3, NIC** is selected and press **Enter**.
 - ii. Press **Tab** twice and then **Enter** to **confirm**.
4. **Provide power to the modem:**
- a. Press **Alt-I** and then Tab to **Cables** and press **Enter**.

- b. Tab to **Power Adapter, AC to DC** and press **Enter**. Press **Enter** again to **add the cable**.
- c. On the **Select Location to Add Item** window:
 - i. In the drop-down menu under **Connector End to Connect**, press the **Down arrow** key until **DC Power Connector** is shown.
 - ii. Press **Tab** to access the **Destination** drop-down menu.
 - iii. Use the **Down arrow** key until **Power Port Fiber Modem** is shown and press **Enter**.
 - iv. Press **Tab** twice and then **Enter** to **Confirm**.
- d. Press **Alt-W** to access the **Workspace**.
- e. Tab until the arrow next to **Fiber Modem/Router** is highlighted and press **Enter**.
- f. Tab six times so the arrow next to **DC Power Connector** is highlighted and press **Enter**.
- g. Tab one time so **Connector, Power, AC Adapter, v2015** is highlighted and press **Enter**. Press **Enter** again to **add the connector**.
- h. In the **Select Location to Add Item Window**:
 - i. Tab to the **Destination** field and press **Enter**. Use the **down arrow** key until **AC3 Power Strip, AC, 115 volt** is selected and press **Enter**.
 - ii. Press **Tab** twice and then **Enter** to **confirm**.

Copyright © 2026 CompTIA, Inc. All rights reserved.